

```
using Optimization, OptimizationOptimJL, Zygote
```

```
f(x)=x[1]^2-2*x[1]+0.6;
```

```
x0=1.8*ones(1)
```

```
optf = OptimizationFunction((x,p)->f(x), ADTypes.AutoZygote())
```

```
prob = OptimizationProblem(optf, x0)
```

```
sol = solve(prob, Optim.Newton(); maxiters=20)
```

Define function to optimize and its derivative (more later)

Define problem and tell the program the initial guess

Find the minimum

```
[ Warning: Verbosity toggle: missing_second_order_ad
  The selected optimization algorithm requires second order derivatives, but `SecondOrder` ADtype was not provided. So
  a `SecondOrder` with AutoForwardDiff() for both inner and outer will be created, this can be suboptimal and not work i
  n some cases so an explicit `SecondOrder` ADtype is recommended.
  @ OptimizationBase ~/.julia/packages/OptimizationBase/EAGYd/src/cache.jl:73
retcode: Success
u: 1-element Vector{Float64}:
 1.0
```

Can change to

Optim.BFGS()
Optim.LBFGS()

